

# Curriculum vitae

## Personal Information

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|----------------------|------------------------------|
| FIRST NAME / SURNAME | Dr. Tatiana Morozova         |
| DATE OF BIRTH        | 24.12.1992                   |
| NATIONALITY          | Russian                      |
| CIVIL STATUS         | married                      |
| EMAIL                | tatiana.morozova@ens-lyon.fr |

## Education

|                       |                                                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| DATES                 | Mar. 2016 - Oct. 2019                                                                                                                |
| QUALIFICATION PROGRAM | <b>Ph.D. in Physics.</b> Thesis on “Polymers and Polymer Colloids in Solvent Mixtures”, under the supervision of Dr. A. Nikoubashman |
| INSTITUTION           | Johannes Gutenberg University, Mainz, <b>Germany</b>                                                                                 |

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|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DATES                 | Sept. 2010 - Jan. 2016                                                                                                                                                                                                                                                   |
| QUALIFICATION AWARDED | <b>Specialist in Physics</b> (higher education program in Condensed Matter Physics equaled a Master's Degree). Thesis on “Computer Simulation of Mechanical Properties of Polyelectrolyte Micro- and Nanocapsules”, under the supervision of Prof. Dr. O. I. Vinogradova |
| INSTITUTION           | Lomonosov Moscow State University, Moscow, <b>Russia</b>                                                                                                                                                                                                                 |

|                       |                                  |
|-----------------------|----------------------------------|
| DATES                 | Sept. 1999 - Jun. 2010           |
| QUALIFICATION AWARDED | <b>High school Diploma</b>       |
| INSTITUTION           | Gymnasium № 1519, Moscow, Russia |

## Work Experience

|             |                                                           |
|-------------|-----------------------------------------------------------|
| DATES       | Dec. 2024- <i>present</i>                                 |
| POSITION    | <b>Permanent</b> researcher (Chargée de Recherche)        |
| INSTITUTION | CNRS, Physics Laboratory at ENS Lyon, Lyon, <b>France</b> |

|             |                                              |
|-------------|----------------------------------------------|
| DATES       | Dec. 2023 - Nov. 2024                        |
| POSITION    | <b>Postdoctoral</b> researcher               |
| INSTITUTION | CNRS, Institut Pasteur, Paris, <b>France</b> |

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|-------------|----------------------------------------------------------------------|
| DATES       | Oct. 2020 - Nov. 2023                                                |
| POSITION    | <b>Postdoctoral</b> researcher                                       |
| INSTITUTION | Institut Max von Laue - Paul Langevin (ILL), Grenoble, <b>France</b> |

|             |                                                                 |
|-------------|-----------------------------------------------------------------|
| DATES       | Nov. 2019 - Sept. 2020                                          |
| POSITION    | <b>Postdoctoral</b> researcher                                  |
| INSTITUTION | LIPhy, CNRS, University Grenoble Alpes, Grenoble, <b>France</b> |

|             |                                                      |
|-------------|------------------------------------------------------|
| DATES       | Mar. 2016 - Oct. 2019                                |
| POSITION    | <b>Ph.D.</b> student                                 |
| INSTITUTION | Johannes Gutenberg University, Mainz, <b>Germany</b> |

## Teaching Experience

|             |                                                                          |
|-------------|--------------------------------------------------------------------------|
| DATES       | Winter semester 2017/2018 & Summer semester 2018                         |
| POSITION    | <b>Teaching assistant</b> for the course “Mathematics for Physicists”,   |
| INSTITUTION | Johannes Gutenberg University, Mainz, Germany                            |
| DATES       | Summer semester 2016                                                     |
| POSITION    | <b>Teaching assistant</b> for the course “Advanced Statistical Physics”, |
| INSTITUTION | Johannes Gutenberg University, Mainz, Germany                            |

## Fundings Raised

|       |                                                                                                                                                                                                                                                                       |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DATES | 2025-2029                                                                                                                                                                                                                                                             |
| NAME  | <b>Jeunes Chercheuses et Jeunes Chercheurs</b> (JCJC) starting grant for young scientists based in France (ANR): “In silico: Disordered Proteins in Cells and Bio-Inspired Materials” (€ 241,000).                                                                    |
| DATES | 2025                                                                                                                                                                                                                                                                  |
| NAME  | <b>CNRS DIALOG</b> : computational equipment (€ 46,000)                                                                                                                                                                                                               |
| DATES | 2024                                                                                                                                                                                                                                                                  |
| NAME  | <b>Marie Skłodowska-Curie Action</b> (MSCA) European Postdoctoral Fellowships, project: “In silico modeling of multi-domain proteins in biological condensates” (231,000€). <i>Declined</i> for the permanent research position at CNRS.                              |
| DATES | 2022 - 2025                                                                                                                                                                                                                                                           |
| NAME  | Franco( <b>ANR</b> )-German( <b>DFG</b> ) project: “Dynamics, kinetics and assembly of model intrinsically disordered proteins from a polymer physics perspective” in collaboration with F. Schreiber, M. Oettel, J.-L. Barrat, O. Matsarskaia, T. Seydel (650,000€). |
| DATES | 2020 - 2026                                                                                                                                                                                                                                                           |
| NAME  | <b>Computational time</b> at the French national cluster facilities IDRIS (CNRS) (300,000 GPU hours granted ~ 110,000€)                                                                                                                                               |
| DATES | 2017                                                                                                                                                                                                                                                                  |
| NAME  | <b>Travel grant</b> from the IRTG/TRR146 graduate school (3,000 €)                                                                                                                                                                                                    |

## Research Visits

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|-------------|-------------------------------------------------------------------------------------------------------------------------------|
| DATES       | July 2022                                                                                                                     |
| PURPOSE     | Collaborative visit to Dr. A. Deblais                                                                                         |
| INSTITUTION | Institute of Physics, University of Amsterdam, <b>Netherlands</b>                                                             |
| DATES       | June - July 2017                                                                                                              |
| PURPOSE     | Experimental project on fabrication of polymer colloids in the research groups of A.Z. Panagiotopoulos, R.K. Prud’homme, R.D. |

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|-------------|------------------------------------------------------------------------------|
|             | Priestley                                                                    |
| INSTITUTION | Department of Chemical and Biological Engineering, Princeton University, USA |

## Skills and Competences

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|---------------------|------------------------------------------------------------------------|
| LANGUAGES SPOKEN    | Russian (native), English (fluent), French (B1), German (intermediate) |
| PROGRAMMING         | C/C++, Python                                                          |
| SIMULATION SOFTWARE | HOOMD-Blue, GROMACS, PyEmma, PLUMED, AlphaFold                         |

## Organization of Scientific Events

|       |                                                                                                                                   |
|-------|-----------------------------------------------------------------------------------------------------------------------------------|
| DATES | Jul. 2021 - Oct. 2023                                                                                                             |
| ROLE  | <b>College secretary</b> of the Theory Group - organization of seminars & group activities, research visits, and workshops at ILL |
| DATES | Feb. - Oct. 2020                                                                                                                  |
| ROLE  | <b>Organizer</b> of group <b>seminars</b> of the Statistical Physics and Modeling team (LIPhy, UGA/CNRS)                          |
| DATES | Oct. 2018 - Oct. 2019                                                                                                             |
| ROLE  | <b>Ph.D. student representative</b> - organization of seminars                                                                    |
| DATES | May 2019                                                                                                                          |
| ROLE  | <b>Organization</b> of the graduate students (CRC TRR146) retreat (5 days, Amsterdam, Netherlands)                                |

## Publications

- (1) **T.I. Morozova** and A. Nikoubashman: Coil-globule collapse of polystyrene chains in tetrahydrofuran-water mixtures, *J. Phys. Chem. B* 122, 2130 (2018)
- (2) **T.I. Morozova**, V.E. Lee, A.Z. Panagiotopoulos, R.K. Prud'homme, R.D. Priestley, and A. Nikoubashman: On the stability of polymeric nanoparticles fabricated through rapid solvent mixing, *Langmuir* 35, 709 (2019)
- (3) **T.I. Morozova** and A. Nikoubashman: Surface activity of soft polymer colloids, *Langmuir* 35, 16907 (2019)
- (4) **T.I. Morozova**, V.E. Lee, N. Bizmark, S.S. Datta, R.K. Prud'homme, A. Nikoubashman, R.D. Priestley: In silico design enables the rapid production of surface-active colloidal amphiphiles, *ACS Cent. Sci.* 6, 166 (2020)
- (5) **T.I. Morozova**, N.A. Garcia, J.-L. Barrat, G.S. Luengo, F. Leonforte: Adsorption and desorption of polymers on bio-inspired chemically structured substrates, *ACS Appl. Mater. Interfaces*, 13, 25, 30086–30097 (2021)
- (6) **T.I. Morozova**, N.A. Garcia, J.-L. Barrat: Temperature

dependence of thermodynamic, dynamical, and dielectric properties of water models, *J. Chem. Phys.*, 156 (12), 126101 (2022)

(7) M.R. Bittermann, **T.I. Morozova**, S.F. Velandia, A. Deblais, S. Woutersen, D. Bonn: Surface-mediated molecular transport in polydisperse oil-in-water emulsions, *Langmuir*, 39, 12, 4207-4215 (2023)

(8) **T.I. Morozova**, N.A. Garcia, O. Matsarskaia, F. Roosen-Runge, J.-L. Barrat: Structural and Dynamical Properties of Elastin-Like Peptides near Lower Critical Solution Temperature, *Biomacromolecules* 24, 4, 1912-1923 (2023) - **top twelve publications** by researchers at the ILL (ILL Highlights 2023)

(9) I. Adroher-Benítez, **T.I. Morozova**, N.A. Garcia, J.-L. Barrat, G.S. Luengo, F. Leonforte: Modeling the adsorption of copolymers on heterogeneous surfaces and the stability of polymer coatings for cosmetic applications, *Macromolecules*, 56, 24, 10285-10295 (2023)

(10) **T.I. Morozova**, N.A. Garcia, J.-L. Barrat: Sequence length controls coil-to-globule transition in elastin-like polypeptides, *J. Phys. Chem. Lett.*, 15, 10757-10762 (2024)

(11) S. Chakraborty, **T.I. Morozova**, J.-L. Barrat: Intrinsically disordered proteins can behave as different polymers across their conformational ensemble, *J. Phys. Chem. B*, 129, 2359-2369 (2025)

## Preprints

(i) V. Schnapka, **T. I. Morozova**, S. Sen, and M. Bonomi, "Atomic resolution ensembles of intrinsically disordered and multi-domain proteins with AlphaFold", (2025)

(ii) S. Sen, S. E. Hoff, **T. I. Morozova**, V. Schnapka, and M. Bonomi, "Advancing in silico drug design with Bayesian refinement of AlphaFold models", (2025)

## Talks and Posters

**Seminar:** Laboratoire de Physique ENS de Lyon, Lyon, 2023: "Thermoresponsive polypeptides: insights from the polymer physics"

**Seminar:** Laboratoire de biochimie théorique, IBPC, Paris, 2023: "Phase behavior of thermo-responsive polypeptides"

**Seminar:** ICS Strasbourg, Strasbourg, 2022, *Structural and Dynamical Properties of Elastin-Like Peptides near Lower Critical Solution Temperature*

**Seminar:** Thematics School in Soft Condensed Matter (Univ, Grenoble Alpes) 2021, *Fabrication of Polymer Colloids through Rapid Solvent Exchange*

**Seminar:** University of Pennsylvania 2019, *Computer simulations of controlled fabrication of polymer colloids through the rapid solvent exchange*

**Invited Talk:** conference FisMat 2025 (Venice, Italy): *Disordered proteins: insights from polymer physics*

**Contributed Talk:** CECAM workshop: Frontiers of Coarse-Grained Models: From New Developments to Modeling Dynamics, Assemblies, and Macromolecular Machines, Lyon (France), **2024**: *Thermo-responsive polypeptides: insights from polymer physics*

**Contributed Talk:** 28th International Conference on Statistical Physics (Statphys28), Tokyo (Japan), **2023**: *Coil-to-globule transition of thermoresponsive polypeptides*

**Contributed Talk:** Spring meeting of the American Chemical Society (online) **2021**, *Polymer adsorption on bio-inspired disordered substrate*

**Contributed Talk:** Spring meeting of the DPG, Regensburg (Germany) **2019**, *Fabrication of polymeric Janus nanoparticles and their behavior at the liquid-liquid interfaces - a simulation study*

**Contributed Talk:** Spring meeting of the DPG Berlin (Germany) **2018**, *Polymer self-assembly into nanoparticles through rapid solvent exchange in organic media*

**Contributed Talk:** 91st ACS Colloid&Surface Science Symposium, New York City (USA) **2017**, *Atomistic simulations of the coil-globule transition of polystyrene chains in solution*

**Contributed Talk:** Spring meeting of the DPG, Dresden (Germany) **2017**, *Controlled fabrication of nanoparticles through rapid solvent exchange*

**Poster:** CECAM workshop: Development of Coarse-Grained Models, Lyon (France) **2022**, *Adsorption and desorption of polymers on bio-inspired chemically heterogeneous surfaces*

**Poster:** Les Houches-TSRC Protein Dynamics Workshop, Aussois (France) **2022**, *Structural and dynamical properties of elastin-like peptides near lower critical solution temperature*

**Poster:** CECAM workshop: From disordered biomolecular complexes to biological coacervates, Zurich (Switzerland) **2022**, *Structural and dynamical properties of elastin-like peptides near lower critical solution temperature*

**Poster:** 5th Edwards Symposium, Cambridge (UK/online) **2021**, *Adsorption and desorption of polymers on bio-inspired chemically heterogeneous surfaces - poster prize*

**Poster:** 3rd Edwards Symposium, Cambridge (UK) **2018**, *Fabrication of Janus nanoparticles from the self-assembly of homopolymers and amphiphiles through rapid precipitation*

**Poster:** CECAM workshop "Collective behavior of soft and active matter under confinement", Mainz (Germany) **2018**, *Fabrication of Janus nanoparticles from the self-assembly of homopolymers and amphiphiles through rapid precipitation*